

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: Power BI And Tableau Practical PRACTICAL - 8

Aim: Performing Advanced Analytics using Trends, Clustering, and Forecasting

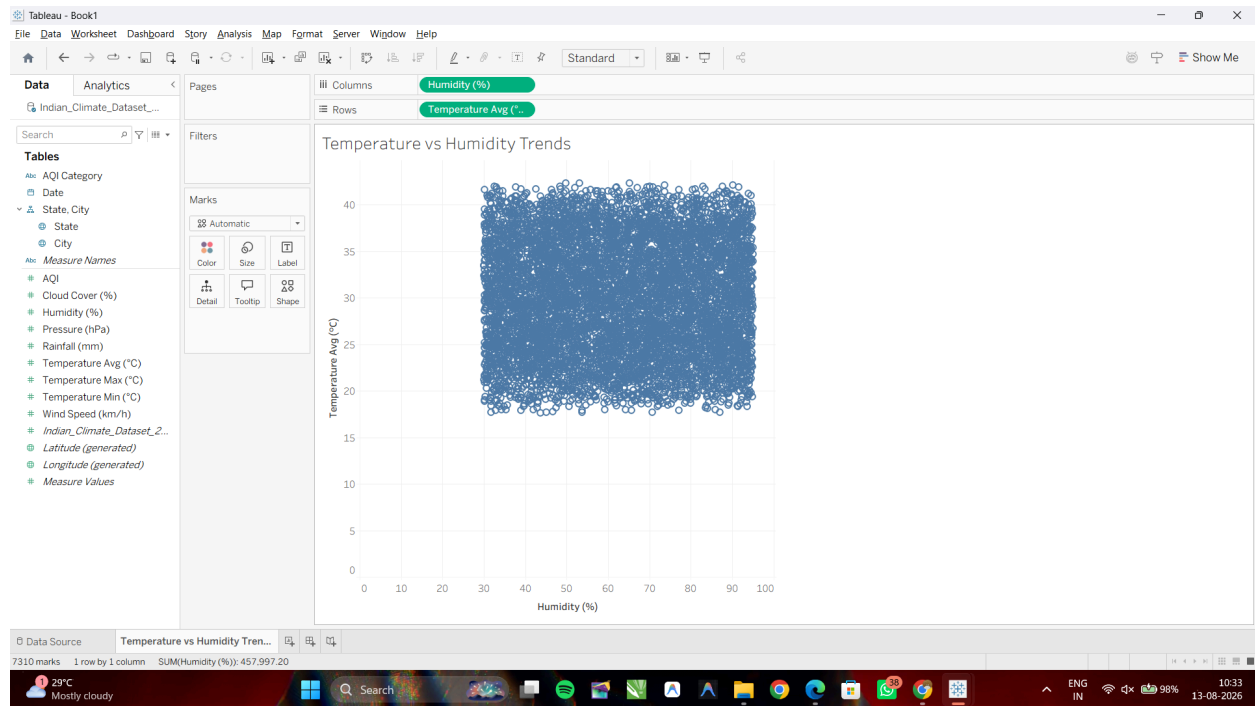
- a) Add and interpret trend lines (linear, exponential)
- b) Customise and analyse trend models
- c) Perform clustering on datasets
- d) Analyse distributions using built-in tools
- e) Apply forecasting techniques
- f) Compare different statistical models
- g) Export and interpret statistical results

Dataset: Indian_Climate_Dataset_2024_2025.csv

Steps:-

A: Add and Interpret Trend Lines (Linear, Exponential)

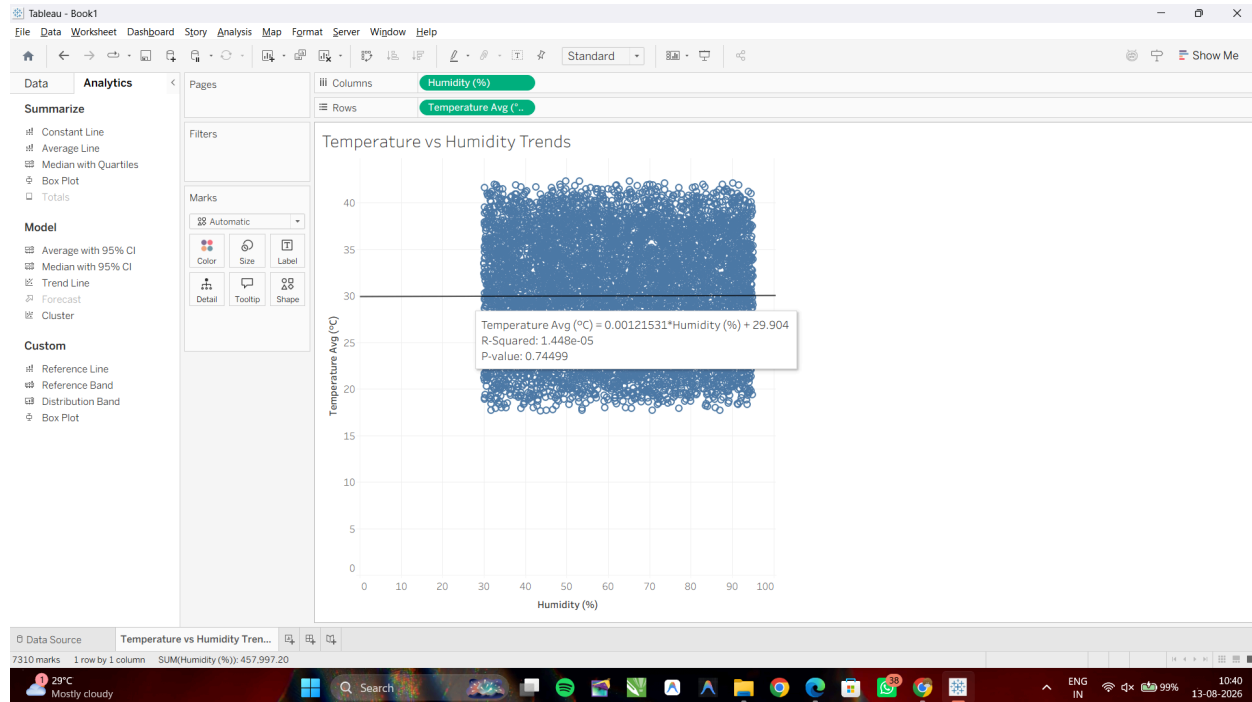
Task 1: Connect Dataset & Build a Scatter Plot / Line Chart



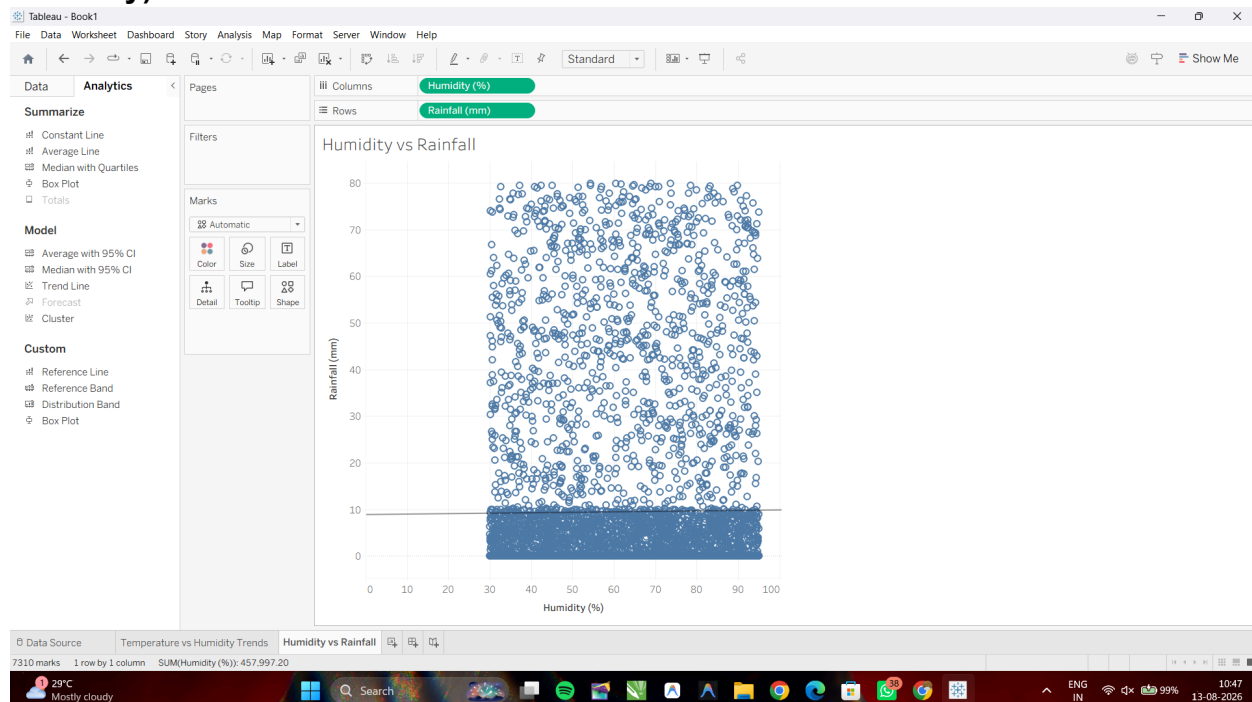
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Task 2: Add a Linear Trend Line.



Task 3: Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)



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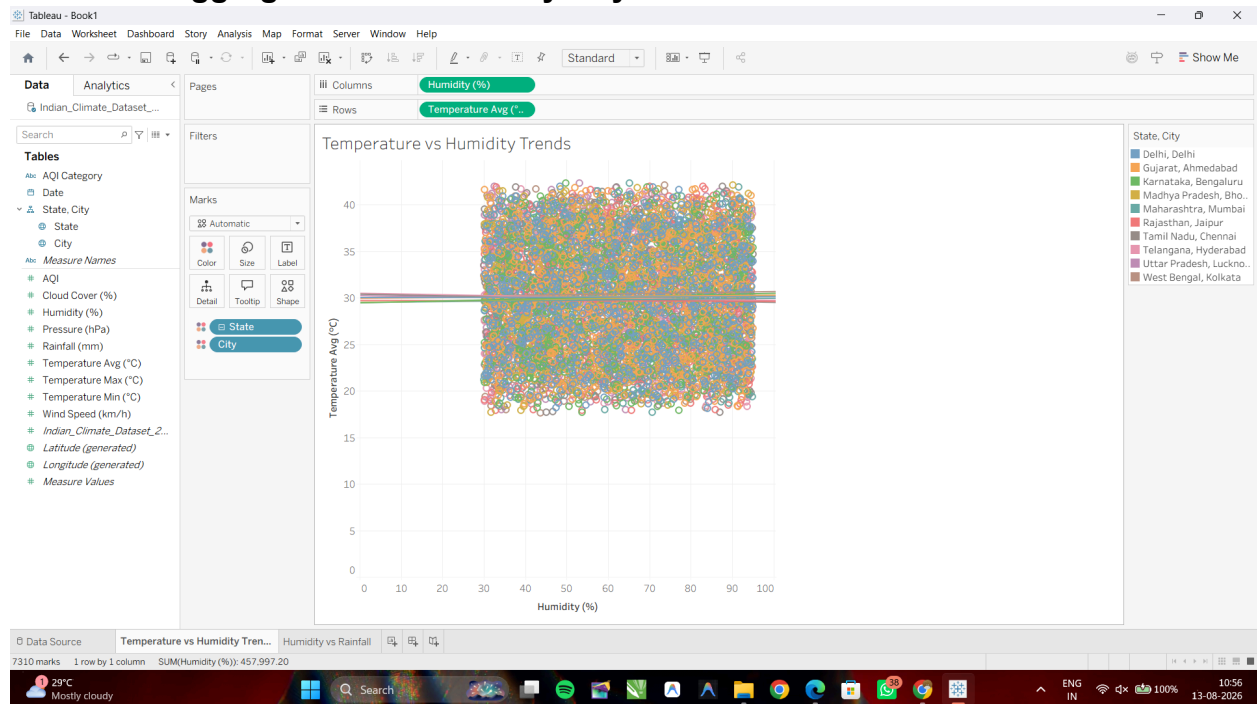
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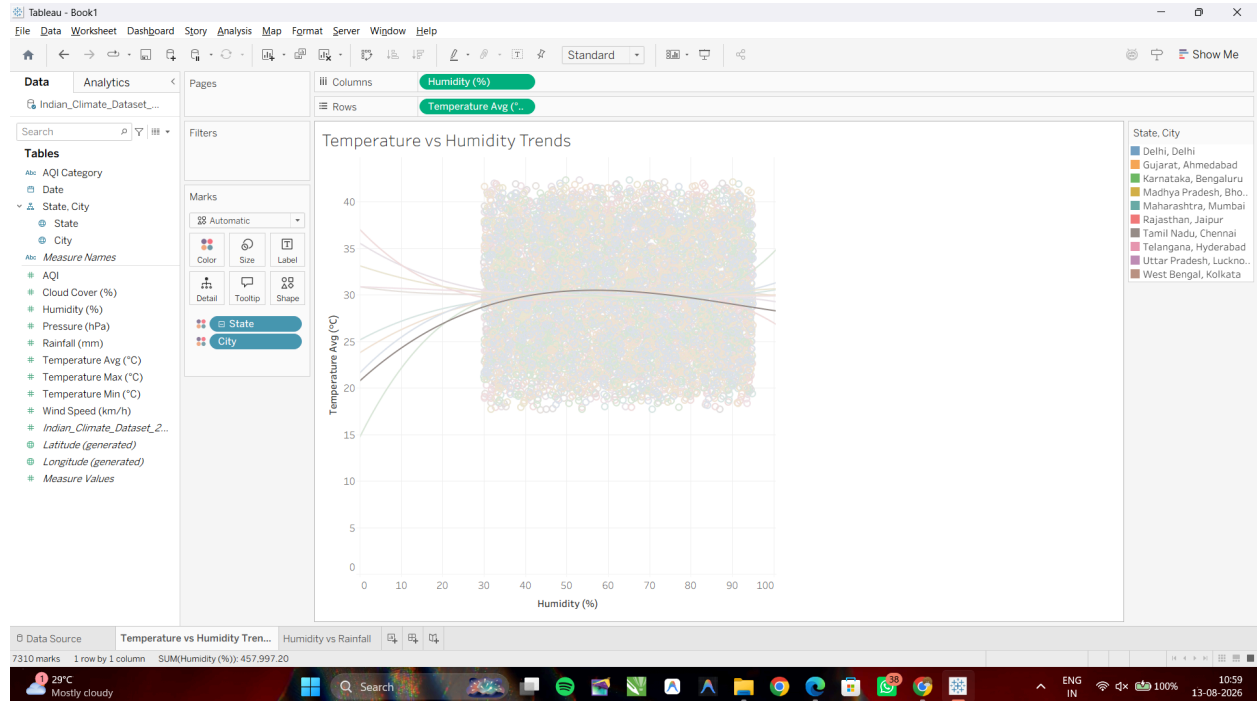
Task 4: Interpret the Results

B: Customize and Analyze Trend Models

Task 1: Disaggregate Trend Lines by City



Task 2: Compare Different Model Types (Polynomial & Logarithmic)



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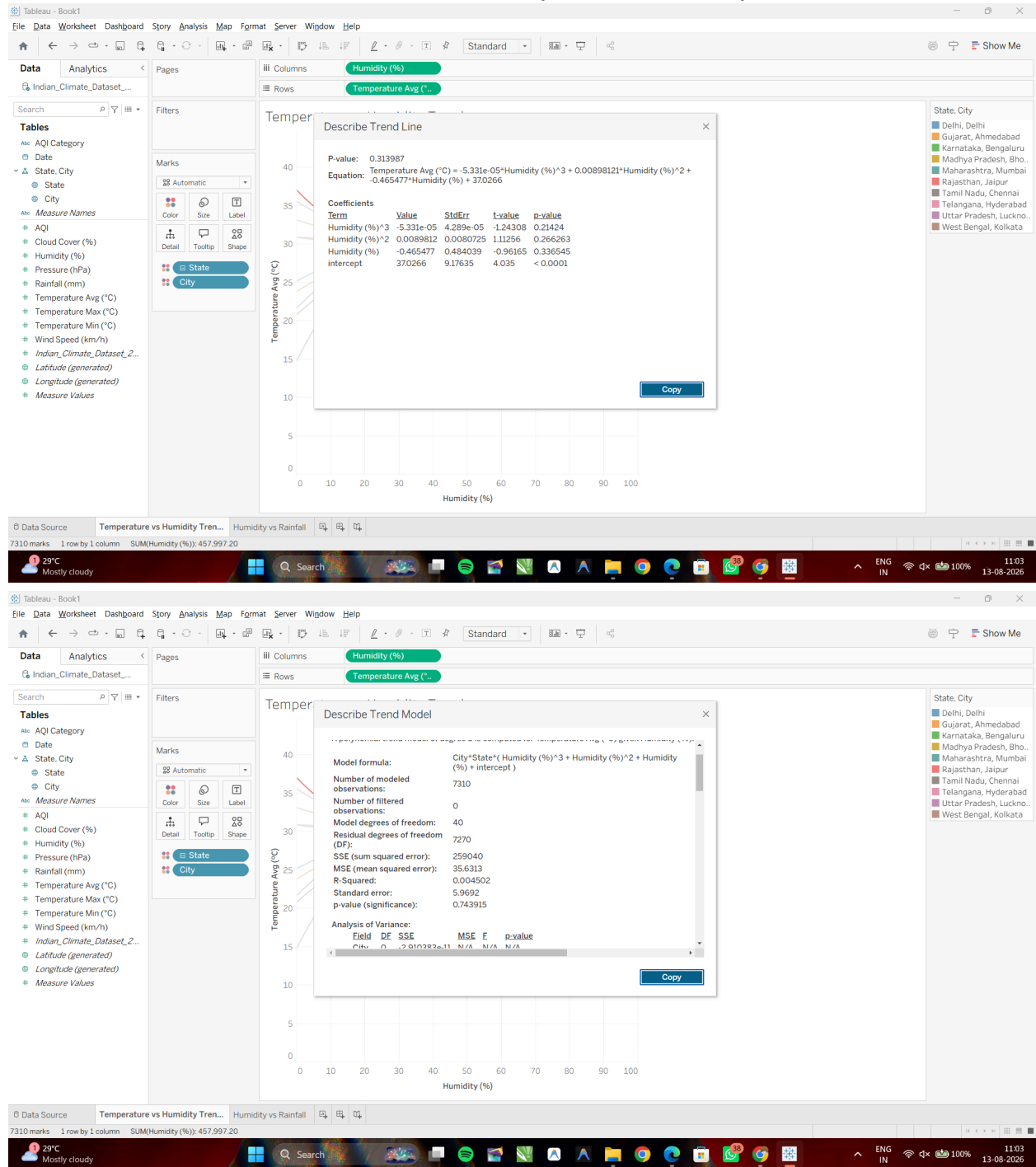
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Task 3: Access Full Statistical Summaries (Describe Model)



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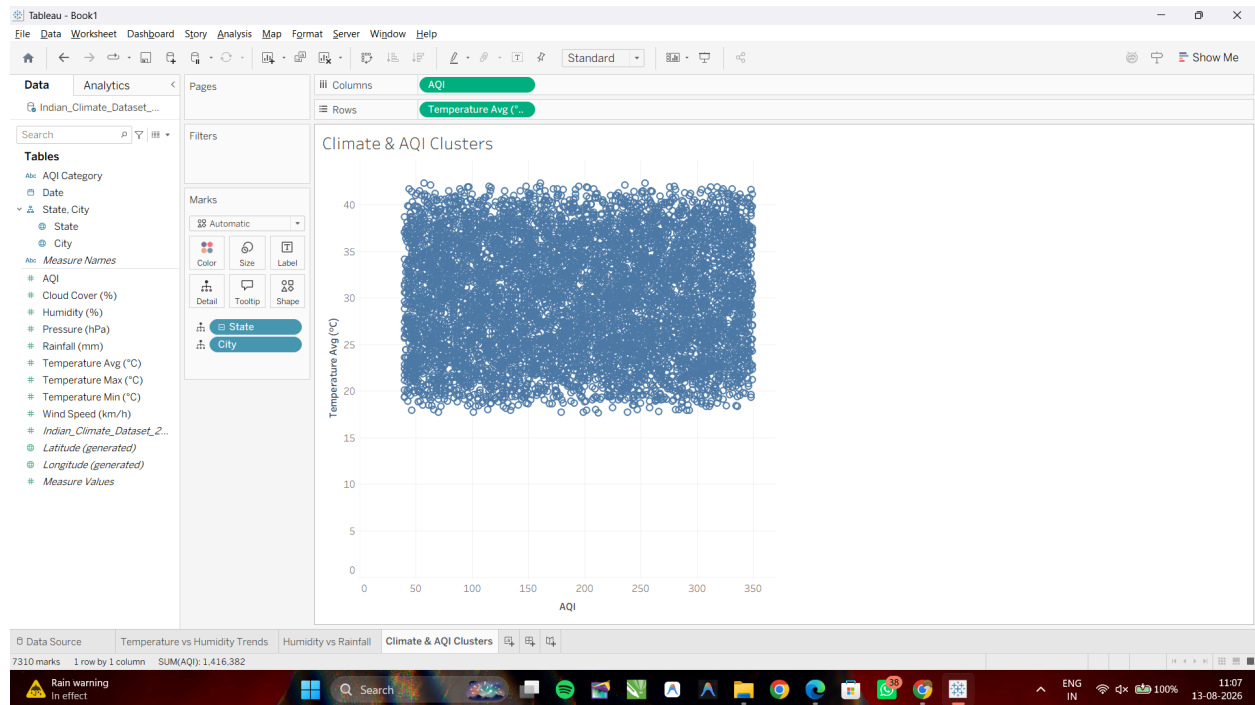
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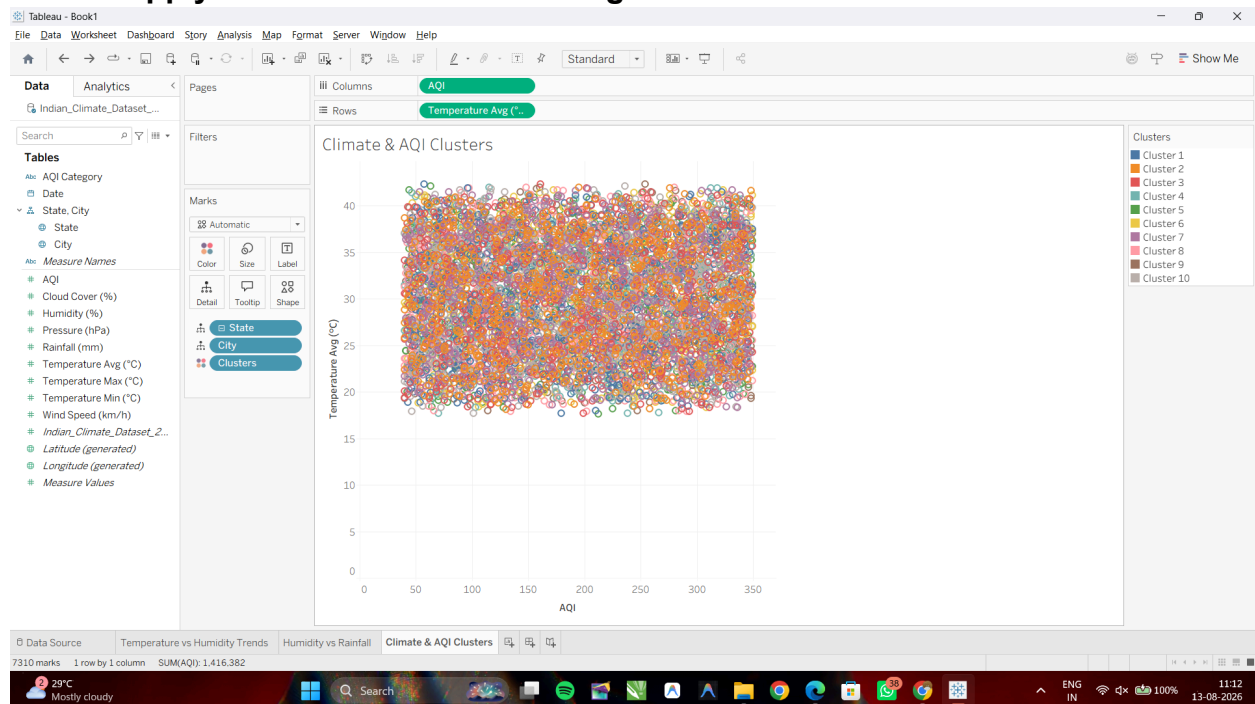
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C: Perform Clustering on Datasets

Task 1: Build a Multi-Metric Scatter Plot Matrix



Task 2: Apply Built-in Tableau Clustering



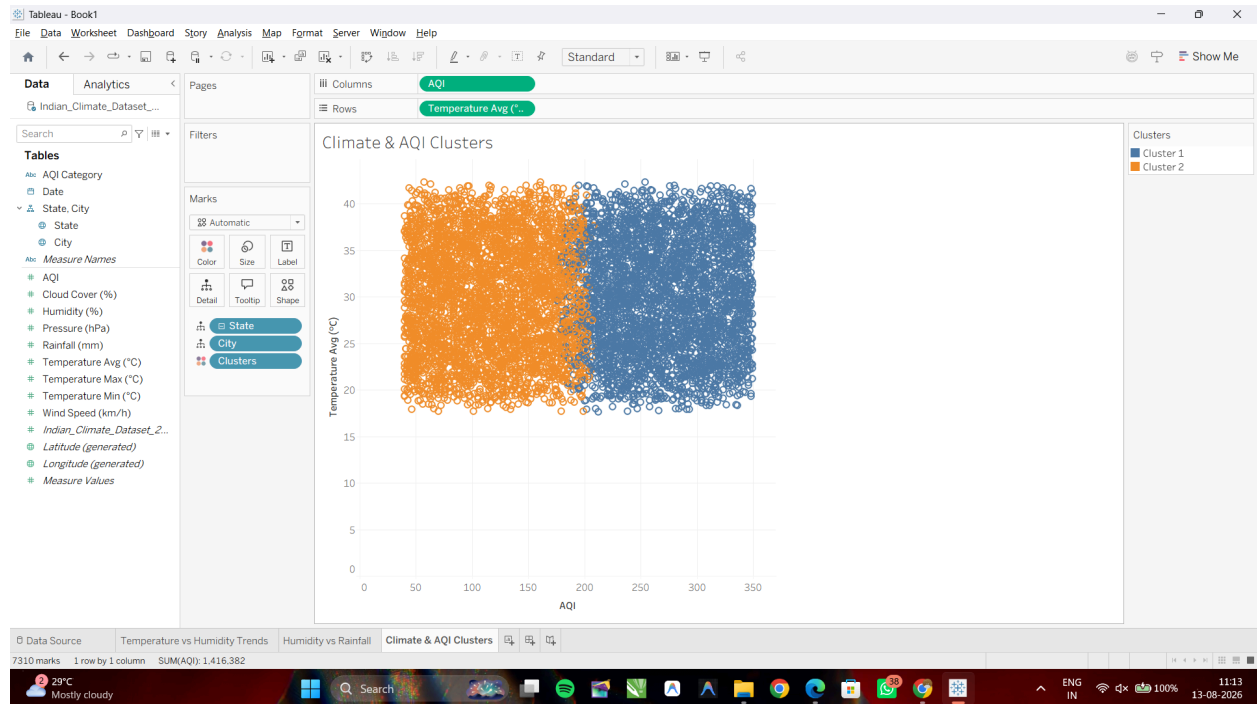
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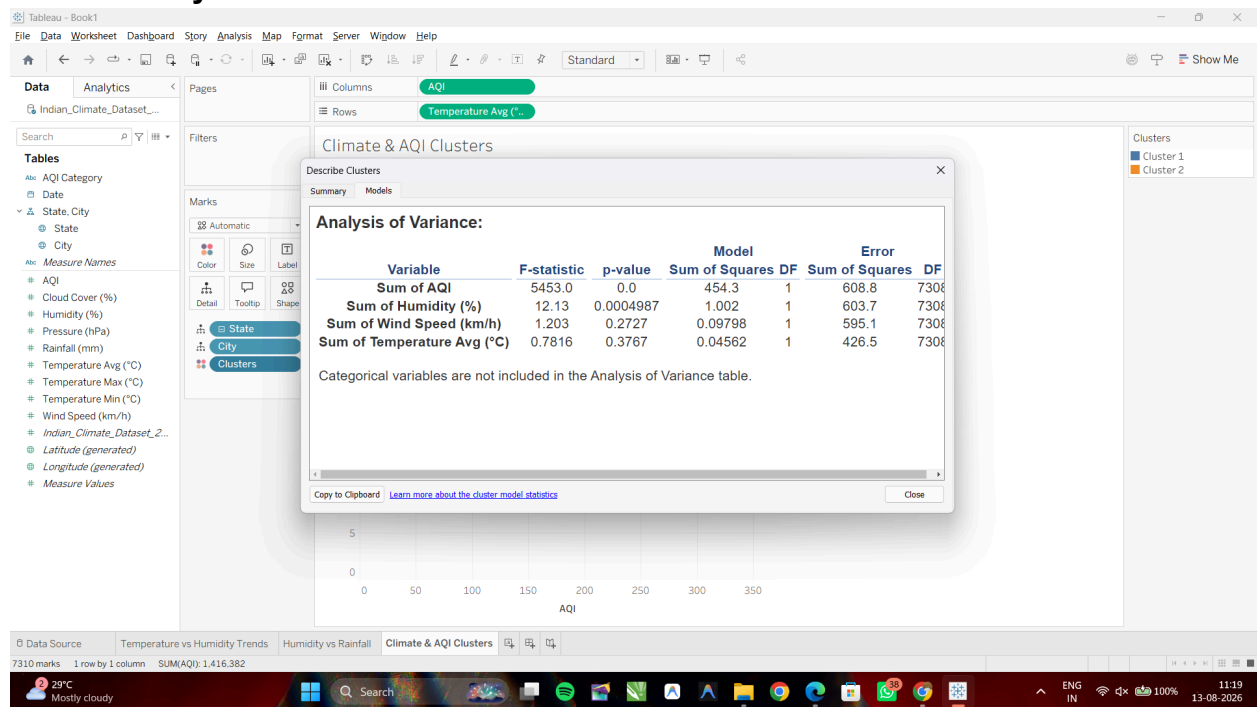
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Task 3: Analyze Cluster Profiles & Save as Dimension



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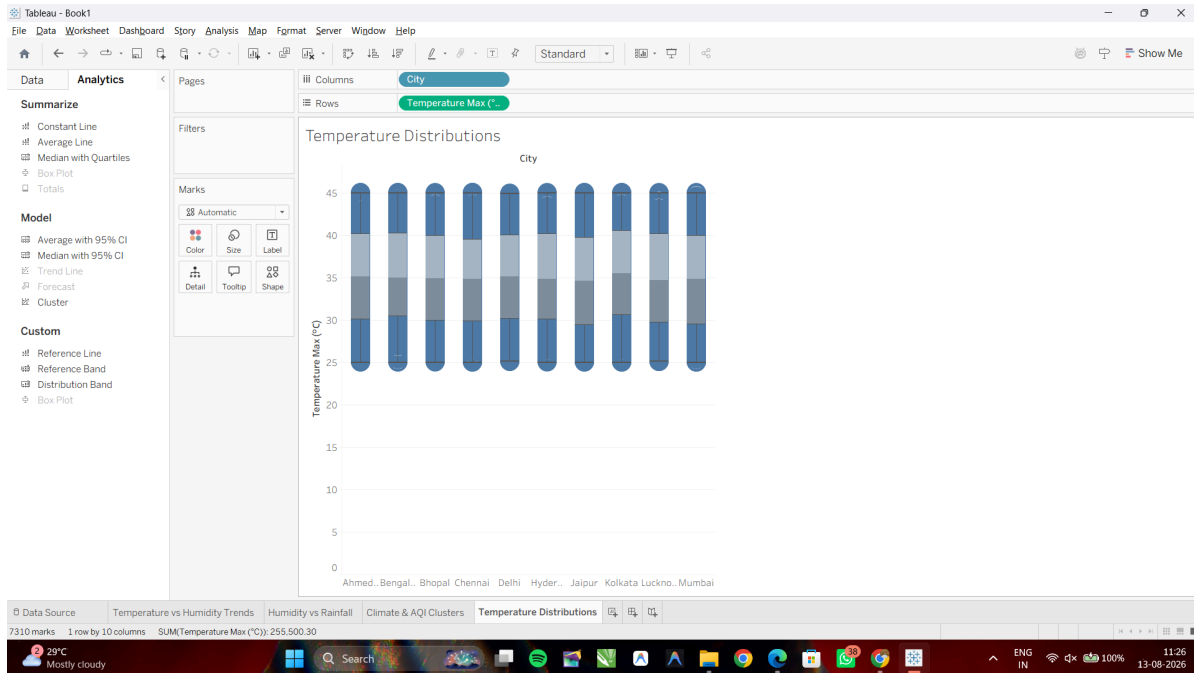
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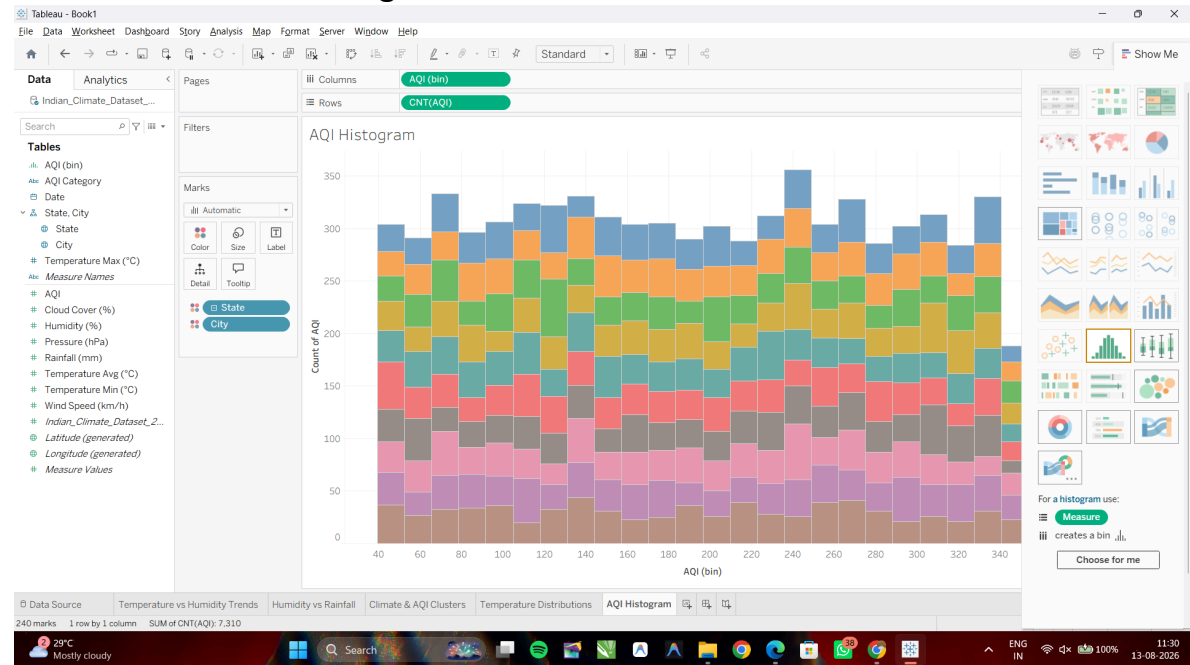
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D: Analyze Distributions using Built-in Tools

Task 1: Create a Box Plot for Temperature Distributions by City.



Task 2: Build an AQI Histogram & Quantile/Percentile View



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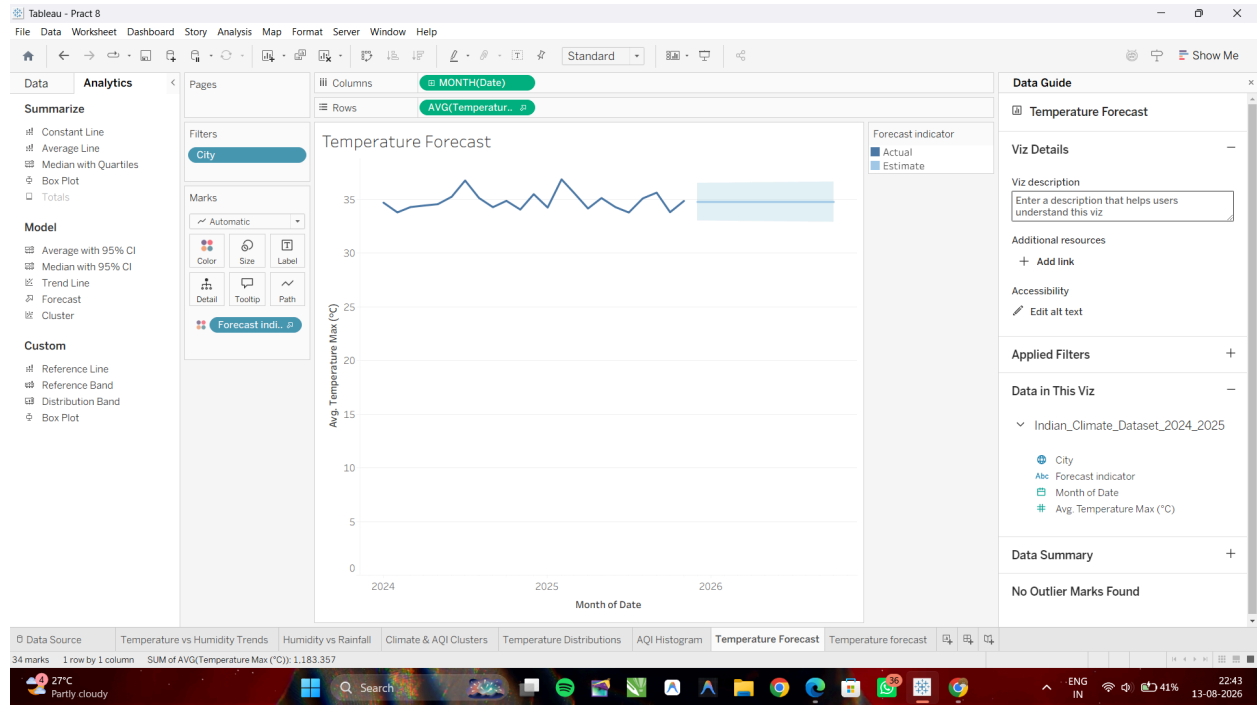
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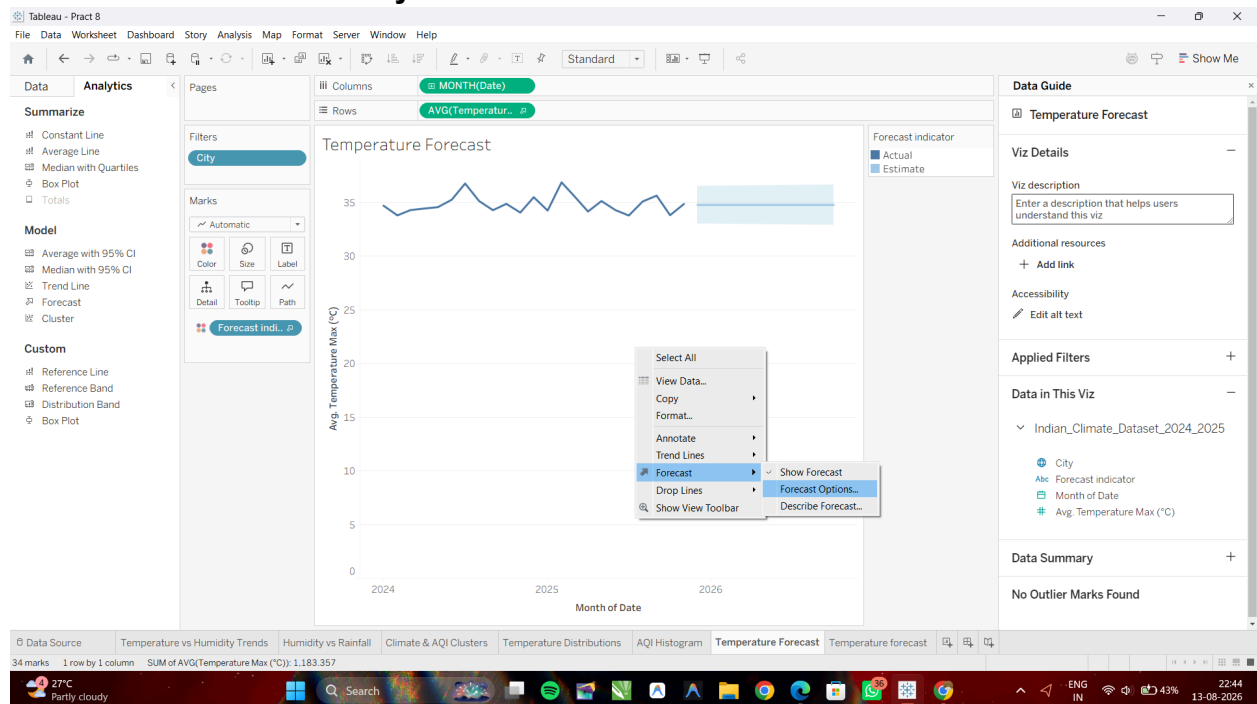
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E: Apply Forecasting Techniques

Task 1: Set Up a Time Series Line Chart

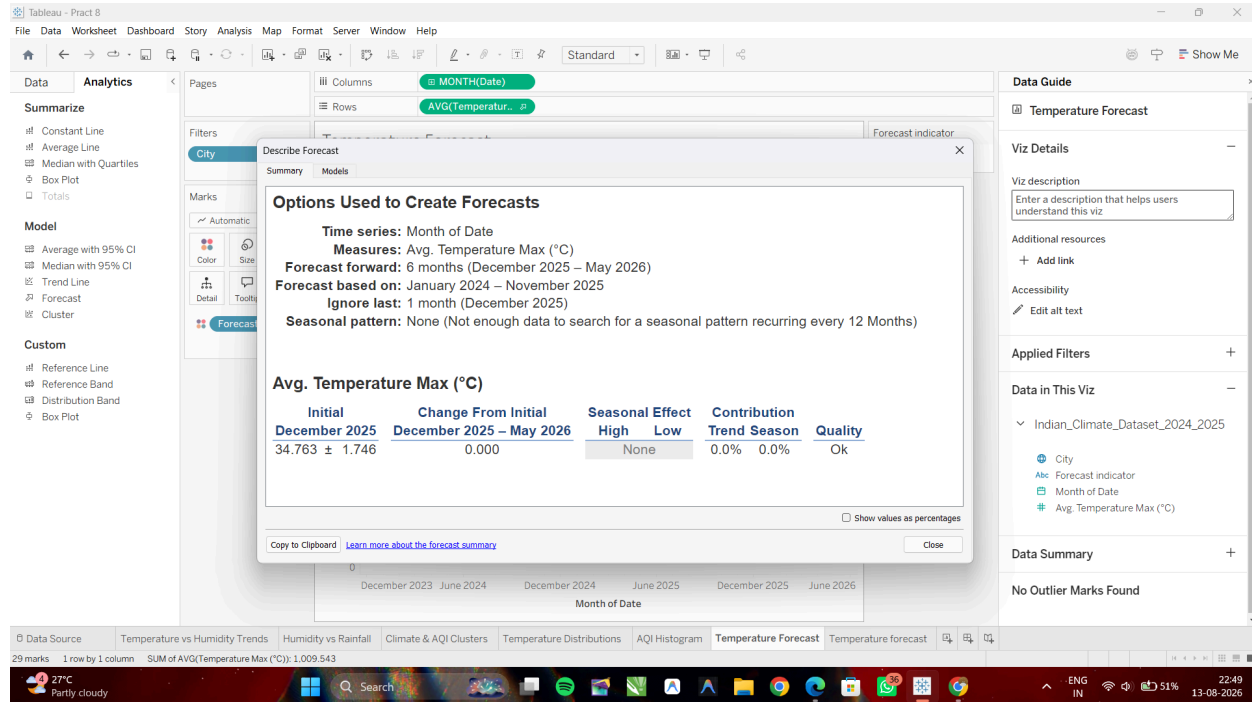


Task 2: Customize & Analyze the Forecast



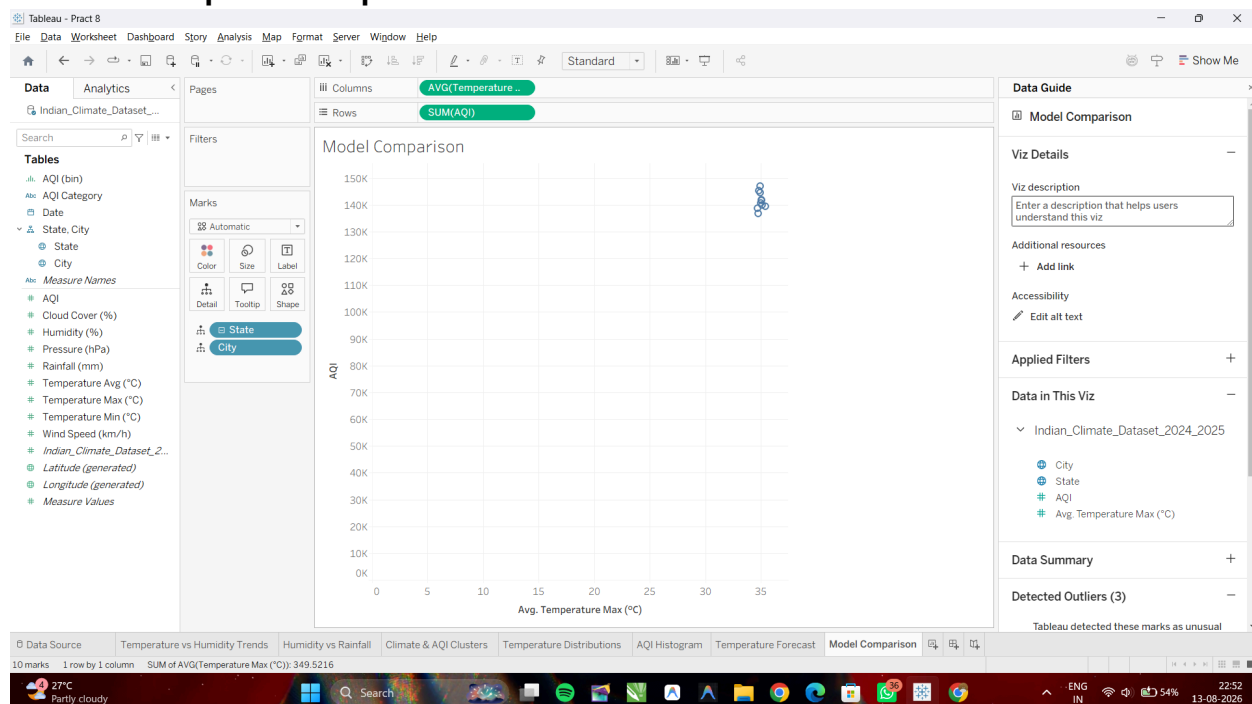
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F: Compare Different Statistical Models

Task 1: Set Up the Comparison View



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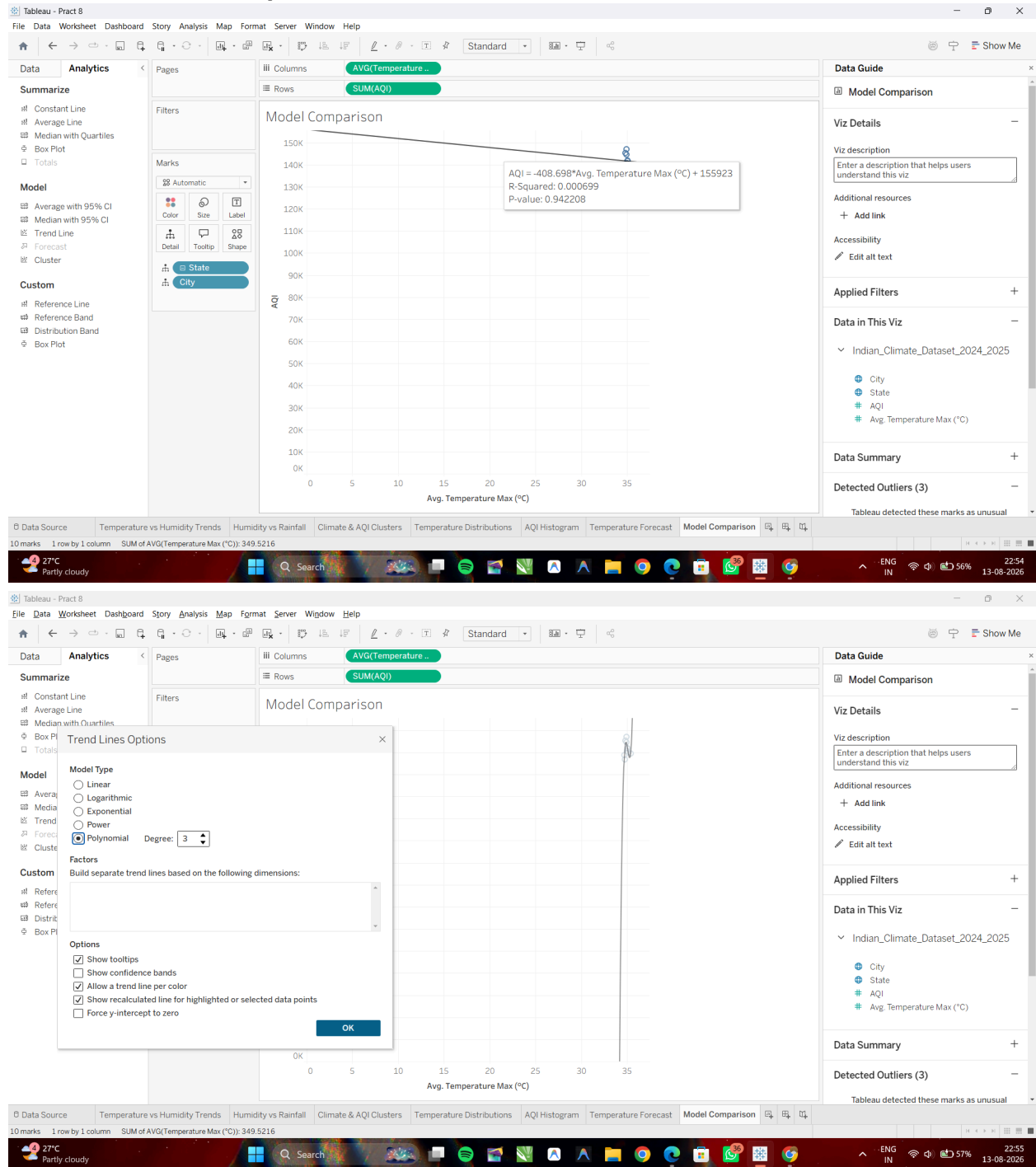
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Task 2: Add and Swap Statistical Models



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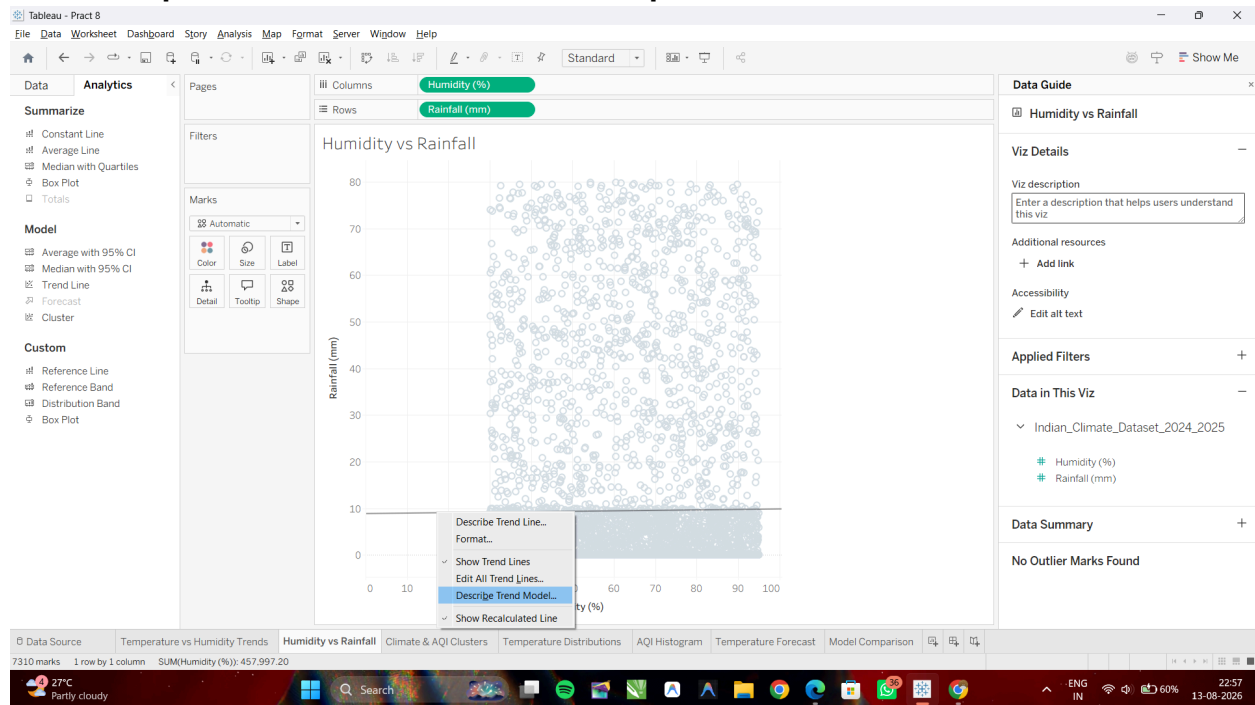
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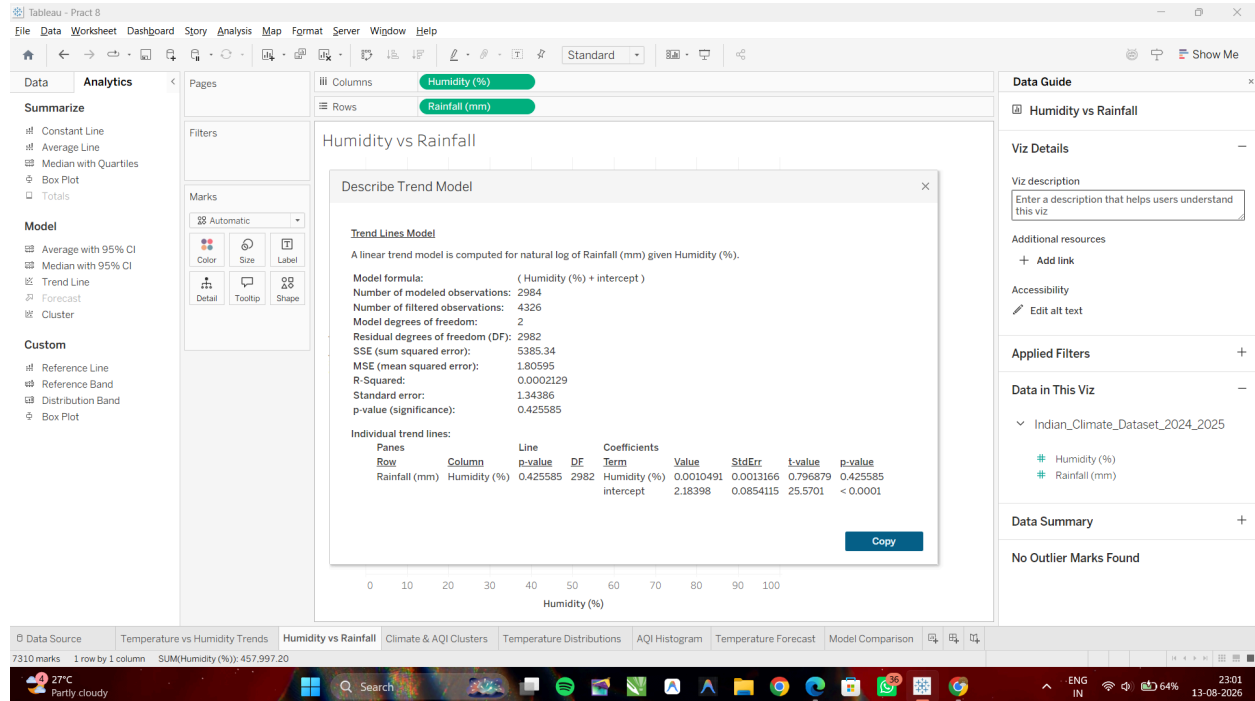
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G: Export and Interpret Statistical Results

Task 1: Export the Statistical Model Description.



Task 2: Interpret the Key Output Metrics



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